

Lecture 3:

What's in a qubit?

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What is a qubit?

A **qubit** is a quantum system represented as a **superposition**, i.e. a linear combination of normalised and mutually orthogonal quantum states labelled by $|0\rangle$ and $|1\rangle$ with **complex** coefficients:

$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

When state $|\phi\rangle$ is **measured** (i.e. **observed**) one of the two basic states $|0\rangle, |1\rangle$ is returned with probability

$$|\alpha|^2 \quad \text{and} \quad |\beta|^2$$

respectively.

What is a qubit?

Being probabilities, the norm squared of coefficients must satisfy

$$|\alpha|^2 + |\beta|^2 = 1$$

which enforces quantum states to be represented by **unit** vectors.

In practice, a qubit is a microscopic system, such as an atom, a nuclear spin, or a polarised photon.

The state space of a qubit

Global phases are useless

Unit vectors equivalent up to multiplication by a complex number of modulus one, i.e. a **phase factor** $e^{i\theta}$, represent the **same** state.

Let $|v\rangle = \alpha|u\rangle + \beta|u'\rangle$ and $e^{i\theta}|v\rangle = e^{i\theta}\alpha|u\rangle + e^{i\theta}\beta|u'\rangle$

$$|e^{i\theta}\alpha|^2 = (\overline{e^{i\theta}\alpha})(e^{i\theta}\alpha) = (e^{-i\theta}\overline{\alpha})(e^{i\theta}\alpha) = \overline{\alpha}\alpha = |\alpha|^2$$

and similarly for β .

As the probabilities $|\alpha|^2$ and $|\beta|^2$ are the **only** measurable quantities, the **global phase has no physical meaning**.

Representation redundancy

qubit state space $\neq$ complex vector space used for representation

The state space of a qubit

Relative phase

It is a measure of the angle between the two complex numbers.
Thus, it **cannot be discarded!**

Those are different states

$$\frac{1}{\sqrt{2}}(|u\rangle + |u'\rangle) \quad \frac{1}{\sqrt{2}}(|u\rangle - |u'\rangle) \quad \frac{1}{\sqrt{2}}(e^{i\theta}|u\rangle + |u'\rangle)$$

...

Going general

Quantum computation explores the laws of quantum theory as computational resources.

Thus, the principles of the former are directly derived from the postulates of the latter.

- The representation postulate
- The evolution postulate
- The composition postulate
- The measurement postulate

Quantum states

The State (or representation) Postulate

The state space of a quantum system is described by a unit vector in a Hilbert space up to a global phase

- In practice, with finite resources, one cannot distinguish between a **continuous** state space from a **discrete** one with arbitrarily small minimum spacing between adjacent locations.
- One may, then, restrict to **finite-dimensional** Hilbert spaces.

The underlying maths is that of Hilbert spaces.

A Language: Dirac's notation

The starting point is a **vector space over the complex field**.

Dirac's bra/ket notation provides a handy (easy to calculate with) way to represent its elements and constructions

$|u\rangle$ A **ket** stands for a vector in an Hilbert space V . In $\mathbb{C}^n$, a column vector of complex entries. The identity for $+$ (the **zero** vector) is just written 0 .

$\langle u|$ A **bra** is a vector in the **dual** space V^* , i.e. scalar-valued linear maps $V \rightarrow \mathbb{C}$ — a row vector in $\mathbb{C}^n$.

There is a bijective correspondence between $|u\rangle$ and $\langle u|$

$$|u\rangle = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \Leftrightarrow [\bar{u}_1 \cdots \bar{u}_n] = \langle u|$$

A Language: Dirac's notation

The bijective correspondence between V and V^* is established by †:

$$(|u\rangle)^\dagger = \langle u| \quad \text{and} \quad (\langle u|)^\dagger = |u\rangle$$

which is **antilinear**:

$$\left(\sum_i \alpha_i |u_i\rangle \right)^\dagger = \sum_i \overline{\alpha_i} \langle u_i|; \quad \text{and} \quad \left(\sum_i \alpha_i \langle u_i| \right)^\dagger = \sum_i \overline{\alpha_i} |u_i\rangle$$

A Language: Dirac's notation

Multiplying a **bra** and **ket** gives a ...

$$\text{a bracket: } \langle u | v \rangle ; \rightsquigarrow \langle u | v \rangle$$

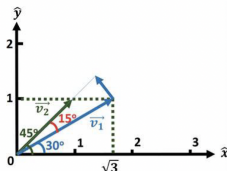
which provides the remaining link in our underlying mathematics:

the **inner product**

$$\text{Notation: } \langle u | v \rangle \equiv \langle u, v \rangle \equiv (|u\rangle, |v\rangle)$$

Recall: what is a inner product?

From a linear algebra textbook: for $\vec{v}_1 = \sqrt{3}\hat{x} + \hat{y}$ and $\vec{v}_2 = \hat{x} + \hat{y}$, the **inner** (or **dot**, or **scalar**) product is



$$\langle \vec{v}_1, \vec{v}_2 \rangle = |\vec{v}_1| |\vec{v}_2| \cos 15 \approx 2.732$$

i.e. the product of the (magnitudes of) $\vec{v}_2$ by $\vec{v}_1$ projected in the direction of $\vec{v}_2$.

The inner product measures **how much the two vectors have in common after scaled by their magnitudes**.

The underlying maths: Hilbert spaces

Complex, inner-product vector space

A complex vector space with **inner product**

$$\langle - | - \rangle : V \times V \longrightarrow \mathbb{C}$$

such that

$$(1) \quad \langle v | \sum_i \lambda_i \cdot |w_i\rangle \rangle = \sum_i \lambda_i \langle v | w_i \rangle$$

$$(2) \quad \langle v | w \rangle = \overline{\langle w | v \rangle}$$

$$(3) \quad \langle v | v \rangle \geq 0 \quad (\text{with equality iff } |v\rangle = 0)$$

Note: $\langle - | - \rangle$ is **conjugate linear** in the first argument:

$$\langle \sum_i \lambda_i \cdot |w_i\rangle | v \rangle = \sum_i \bar{\lambda}_i \langle w_i | v \rangle$$

and **linear** in the second.

Inner product: examples

In $\mathbb{C}$

$$\langle a + bi | c + di \rangle = \overline{(a + bi)}(c + di)$$

In $\mathbb{C}^n$

$$\langle u | v \rangle = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \underbrace{[\overline{u_1} \quad \overline{u_2} \quad \cdots \quad \overline{u_n}]}_{\langle u |} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}}_{| v \rangle} = \sum_{i=1}^n \overline{u_i} v_i$$

where $\bar{c}$ is the complex conjugate of c

The dual space

 V^*

If V is a Hilbert space, V^* is the space of **linear maps** from V to $\mathbb{C}$.

Elements of V^* are denoted by

$$\langle u| : V \longrightarrow \mathbb{C}$$

as discussed above, and defined through the **inner product**:

$$\langle u|(|v\rangle) = \langle u|v\rangle$$

In a matricial representation $\langle u|$ is the **Hermitian conjugate**, or **conjugate transpose** of $|u\rangle$,

i.e. the **transpose** of the vector formed by the **complex conjugate** of each element in $|u\rangle$.

Maps in a Hilbert space

Maps between Hilbert spaces are, of course, **linear transformations** which are typically represented by **matrices** whose entries are computed through the inner product:

$$A_{i,j} = \langle i|A|j\rangle$$

The adjoint map

The adjoint $U^\dagger$ of a map $U : V \longrightarrow V$ is the unique map satisfying

$$(U^\dagger|w\rangle, |v\rangle) = (|w\rangle, U|v\rangle) \quad \text{or, in a simplified notation}$$

$$\langle U^\dagger w | v \rangle = \langle w | Uv \rangle$$

an equality that is often denoted by the expression

$$\langle u | U | v \rangle$$

Concretely,

$$\langle i | U^\dagger | j \rangle = \overline{\langle j | U | i \rangle}$$

i.e., the matrix representation of $U^\dagger$ is the conjugate transpose of U

Properties

- $(UV)^\dagger = V^\dagger U^\dagger$
- $U^{\dagger\dagger} = U$

Other old friends: Norms and orthogonality

Old friends

- $|v\rangle$ and $|w\rangle$ are **orthogonal** if $\langle v|w\rangle = 0$
- **norm**: $\|v\| = \sqrt{\langle v|v\rangle}$, a nonnegative real number

This norm satisfies $\|v\rangle + |w\rangle\| \leq \|v\rangle\| + \|w\rangle\|$ due to the Cauchy-Schwarz inequality:

$$\langle x|y\rangle^2 \leq \langle x|x\rangle \langle y|y\rangle$$

Other old friends: Norms and orthogonality

Recall

- **normalization**: $\frac{|v\rangle}{||v\rangle|}$
- $|v\rangle$ is a **unit vector** if $||v\rangle| = 1$
- A set of vectors $\{|i\rangle, |j\rangle, \dots, \}$ is **orthonormal** if each $|i\rangle$ is a unit vector and

$$\langle i | j \rangle = \delta_{i,j} = \begin{cases} i = j & \Rightarrow 1 \\ \text{otherwise} & \Rightarrow 0 \end{cases}$$

Other old friends: Bases

Orthonormal basis

A orthonormal basis for a Hilbert space V of dimension n is a set $B = \{|i\rangle\}$ of n linearly independent elements of V st

- $\langle i|j\rangle = \delta_{i,j}$ for all $|i\rangle, |j\rangle \in B$
- and B **spans** V , i.e. every $|v\rangle$ in V can be written as

$$|v\rangle = \sum_i \alpha_i |i\rangle \quad \text{for some } \alpha_i \in \mathbb{C}$$

Changing representations of a quantum state from one basis to another is a common technique in quantum algorithms.

(cf *why superposition can help you to date the girl/boy of your choice?*)

Example: The Hadamard basis

One of the infinitely many orthonormal bases for a space of dimension 2:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Check e. g.

$$\langle + | - \rangle = \frac{1}{2}(|0\rangle + |1\rangle, |0\rangle - |1\rangle) = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0$$

$$\|+\rangle\| = \sqrt{\langle + | + \rangle} = \sqrt{\frac{1}{2}(|0\rangle + |1\rangle, |0\rangle + |1\rangle)} = \sqrt{\frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = 1$$

Old friends: Bases

If $|v\rangle$ is expressed wrt an orthonormal basis, i.e. $|v\rangle = \sum_i \alpha_i |i\rangle$, then the **amplitude** of $|v\rangle$ wrt $|i\rangle$ satisfies

$$\alpha_i = \langle i|v\rangle$$

because

$$\begin{aligned}\langle i|v\rangle &= \langle i|\sum_j \alpha_j |j\rangle \\ &= \sum_j \alpha_j \langle i|j\rangle \\ &= \sum_j \alpha_j \delta_{i,j} \\ &= \alpha_i\end{aligned}$$

Hilbert spaces

The complete picture

An **Hilbert space** is an inner-product space V st the metric defined by its norm turns V into a **complete metric space**, i.e.any Cauchy sequence

$$|v_1\rangle, |v_2\rangle, \dots$$

$$\forall_{\epsilon>0} \exists_N \forall_{m,n>N} ||v_m - v_n\rangle| \leq \epsilon$$

converges

(i.e. there exists an element $|s\rangle$ in V st $\forall_{\epsilon>0} \exists_N \forall_{n>N} ||s - v_n\rangle| \leq \epsilon$)

The completeness condition is trivial in **finite dimensional** vector spaces

The evolution postulate

If a quantum state is a **ray** (i.e. a unit vector in a Hilbert space H up to a global phase), its evolution is specified a certain kind of **linear** operators $U: H \longrightarrow H$.

Linearity

$$U \left(\sum_j \alpha_j |v_j\rangle \right) = \sum_j \alpha_j U(|v_j\rangle)$$

Just by itself, linearity has an important consequence:

quantum states cannot be cloned

The no-cloning theorem

Linearity implies that quantum states cannot be cloned

Let $U(|a\rangle|0\rangle) = |a\rangle|a\rangle$ be a 2-qubit operator and $|c\rangle = \frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$ for $|a\rangle, |b\rangle$ orthogonal.

If U is linear, then

$$U\left(\frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)|0\rangle\right) = \frac{1}{\sqrt{2}}(U(|a\rangle|0\rangle) + U(|b\rangle|0\rangle)) = \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |b\rangle|b\rangle)$$

which is different from

$$U(|c\rangle|0\rangle) = |c\rangle|c\rangle = \frac{1}{\sqrt{2}}(|a\rangle|a\rangle + |a\rangle|b\rangle + |b\rangle|a\rangle + |b\rangle|b\rangle)$$

Computing with qubits

The evolution postulate

The evolution over time of the state of a closed quantum system is described by a unitary operator.

The evolution is **linear**

$$U \left(\sum_j \alpha_j |v_j\rangle \right) = \sum_j \alpha_j U(|v_j\rangle)$$

and preserves the **normalization constraint**

$$\text{If } \sum_j \alpha_j U(|v_j\rangle) = \sum_j \alpha'_j |v_j\rangle \text{ then } \sum_j |\alpha'_j|^2 = 1$$

Computing with qubits

Preservation of the **normalization constraint** means that unit length vectors (and thus orthogonal subspaces) are mapped by U to unit length vectors (and thus to orthogonal subspaces).

This entails a condition on valid quantum operators: they must **preserve** the inner product, i.e.

$$(U|v\rangle, U|w\rangle) = \langle v|U^\dagger U|w\rangle = \langle v|w\rangle$$

which is the case iff U is **unitary**, i.e. $U^\dagger = U^{-1}$, because

$$U^\dagger U = UU^\dagger = I$$

Unitarity

- Preserving the inner product means that a unitary operator maps **orthonormal bases** to **orthonormal bases**.
- Conversely, any operator with this property is unitary.
- If given in matrix form, being unitary means that the set of columns of its matrix representation are orthonormal (because the j th column is the image of $U|j\rangle$). Equivalently, rows are orthonormal (why?)

Unitarity

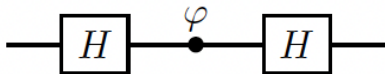
Unitarity is the **only** constraint on quantum operators: Any unitary matrix specifies a valid quantum operator.

This means that there are many non-trivial operators on a single qubit, in contrast with the **classical** case where the only non-trivial operation on a bit is **complement**.

Finally, because the **inverse** of a unitary matrix is also a unitary matrix, a quantum operator can always be inverted by another quantum operator

Unitary transformations are **reversible**

Recall the golden pattern



$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

Hadamard gate

and

$$P_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}$$

Phase shift gate

yielding

$$A = HP_\varphi H = e^{i\frac{\varphi}{2}} \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}$$

Recall the golden pattern

If the product $HP_\varphi H$ describes the action of the whole circuit, one may also step through its execution (ignoring the global phase), as follows

$$\begin{aligned} |0\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ &\xrightarrow{P_\varphi} \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle) \\ &\xrightarrow{H} \left(\cos\left(\frac{\varphi}{2}\right)|0\rangle - i\sin\left(\frac{\varphi}{2}\right)|1\rangle\right) \end{aligned}$$

All the interference is controlled by the phase gate P_φ .

Phase gates: Three remarkable cases

Phase-flip $(\varphi = \pi)$ $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$\frac{\pi}{4}$ -phase $(\varphi = \frac{\pi}{2})$ $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$\frac{\pi}{8}$ -phase $(\varphi = \frac{\pi}{4})$ $T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$

Note that

$$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix} = \begin{bmatrix} e^{-i\frac{\varphi}{2}} & 0 \\ 0 & e^{i\frac{\varphi}{2}} \end{bmatrix}$$

why?

Pauli gates

$$\text{Identity} \quad I = |0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Bit-flip} \quad X = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Phase-flip} \quad Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = P_\pi$$

$$\text{BitPhase-flip} \quad Y = i(-|1\rangle\langle 0| + |0\rangle\langle 1|) = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

The X (bit-flip) gate

The $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ gate



$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

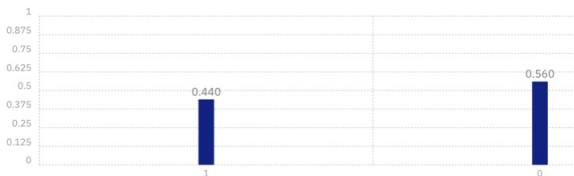
Pauli gates: Properties

Pauli gates

- are unitary and **Hermitian** ($G^\dagger = G$)
- square to the identity
- are anticommutative: $XY = -YX$, $XZ = -ZX$ and $YZ = -ZY$.

The Hadamard gate creates superpositions

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



$$H|0\rangle = |+\rangle = \overbrace{\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)}^{\text{superposition}}$$
$$H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Pauli gates in the golden pattern

A few equalities

$$I = HH$$

$$X = HZH$$

$$Z = HXH$$

$$-Y = HYH$$

States and gates

Quantum gates

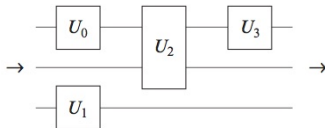
A **gate** is a transformation that acts on only a small number of qubits
Differently from the classical case, they do not necessarily correspond to physical objects

Is there a complete set?

In general no: there are uncountably many quantum transformations, and a finite set of generators can only generate countably many elements.

However, it is possible for finite sets of gates to generate arbitrarily close approximations to all unitary transformations.

Circuits

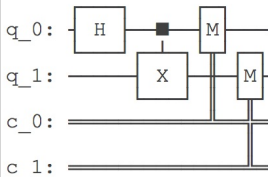


The circuit model

Circuits in Qiskit

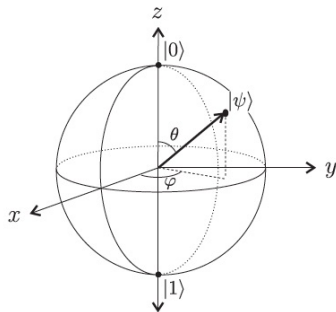
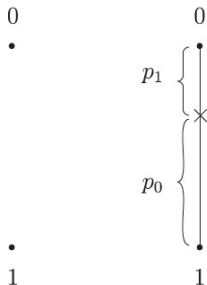
```
from qiskit import QuantumCircuit
```

```
qc = QuantumCircuit(2, 2)  
qc.h(0)  
qc.cx(0, 1)  
qc.measure([0, 1], [0, 1])  
qc.draw()
```



A handy representation for a single qubit

Deterministic, probabilistic and quantum bits



(from [Kaeys et al, 2007])

A handy representation for a single qubit

There is a simple way to visualise single-qubit state vectors. i.e.

$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$$

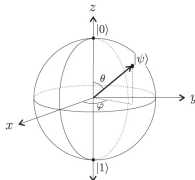
constrained by the relation

$$|\alpha|^2 + |\beta|^2 = 1$$

in terms of Euclidean vectors in three dimensions as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

Inspecting the Bloch sphere



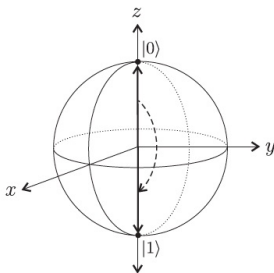
- Poles represent the classical bits. In general, **orthogonal states correspond to antipodal points** and every **diameter** to a **basis** for the single-qubit state space.
- Once measured a qubit collapses to one of the two poles. Which pole depends exactly on the arrow direction. The angle θ measures the collapsing **probability**: If the arrow points at the equator, there is 50-50 chance to collapse to any of the two poles.

Moreover, any unitary transformation on the state vector induces a **rotation** of the corresponding Bloch vector.

Inspecting the Bloch sphere: The X gate

The action of a **1-gate** U on a quantum state $|\phi\rangle$ can be thought of as a **rotation** of the Bloch vector for $|\phi\rangle$ to the Bloch vector for $U|\phi\rangle$, eg.

Example: X



is a rotation about the x axis.

Inspecting the Bloch sphere: The phase shift gate P_ϕ

$$P_\phi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix}$$

$$P_\phi |0\rangle = |0\rangle$$

$$P_\phi |1\rangle = e^{i\phi} |1\rangle$$

The gate acts by

$$\cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \mapsto \cos \frac{\theta}{2} |0\rangle + e^{i(\varphi+\phi)} \sin \frac{\theta}{2} |1\rangle$$

The azimuthal angle changes from φ to $\varphi + \phi$ and so the Bloch sphere is rotated anticlockwise by ϕ about the z -axis.

Note that rotating a vector wrt the z -axis does not affect which state the arrow will collapse to, when measured.

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The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- Express $|\psi\rangle$ in **polar** form

$$|\psi\rangle = \rho_1 e^{i\varphi_1} |0\rangle + \rho_2 e^{i\varphi_2} |1\rangle$$

- Eliminate one of the four real parameters multiplying by $e^{-i\varphi_1}$

$$|\psi\rangle = \rho_1 |0\rangle + \rho_2 e^{i(\varphi_2 - \varphi_1)} |1\rangle = \rho_1 |0\rangle + \rho_2 e^{i\varphi} |1\rangle$$

making $\varphi = \varphi_2 - \varphi_1$,

which is possible because **global phase factors** are **physically meaningless**.

The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- Switching back the coefficient of $|1\rangle$ to Cartesian coordinates

$$|\psi\rangle = \rho_1|0\rangle + (a + bi)|1\rangle$$

the normalization constraint

$$|\rho_1|^2 + |a + ib|^2 = |\rho_1|^2 + (a - ib)(a + ib) = |\rho_1|^2 + a^2 + b^2 = 1$$

yields the **equation of a unit sphere** in the real tridimensional space with Cartesian coordinates: (a, b, ρ_1) .

The Bloch sphere: Representing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

- The **polar** coordinates (ρ, θ, φ) of a point in the surface of a sphere relate to Cartesian ones (x, y, z) through the correspondence

$$x = \rho \sin \theta \cos \varphi$$

$$y = \rho \sin \theta \sin \varphi$$

$$z = \rho \cos \theta$$

- Recalling $\rho = 1$ (cf unit sphere),

$$\begin{aligned} |\psi\rangle &= \rho_1|0\rangle + (a + ib)|1\rangle \\ &= \cos \theta |0\rangle + \sin \theta (\cos \varphi + i \sin \varphi) |1\rangle \\ &= \cos \theta |0\rangle + e^{i\varphi} \sin \theta |1\rangle \end{aligned}$$

which, with **two parameters**, defines a **point** in the sphere's surface.

The Bloch sphere

Actually, one may just focus on the **upper hemisphere** ($0 \leq \theta' \leq \frac{\pi}{2}$) as opposite points in the lower one differ only by a phase factor of -1 , as suggested by

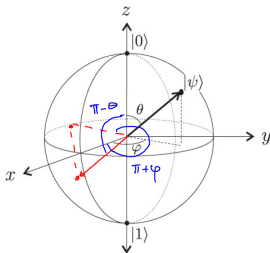
$$\theta' = 0 \Rightarrow |\psi\rangle = \cos 0|0\rangle + e^{i\varphi} \sin 0|1\rangle = |0\rangle$$

$$\theta' = \frac{\pi}{2} \Rightarrow |\psi\rangle = \cos \frac{\pi}{2}|0\rangle + e^{i\varphi} \sin \frac{\pi}{2}|1\rangle = e^{i\varphi}|1\rangle = |1\rangle$$

Note that **longitude** (φ) is irrelevant in a pole!

The Bloch sphere

Indeed, let $|\psi'\rangle$ be the opposite point on the sphere with polar coordinates $(1, \pi - \theta, \varphi + \pi)$:



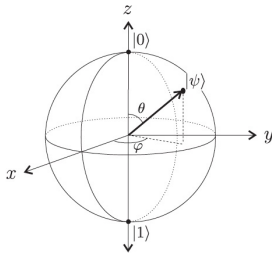
$$\begin{aligned}
 |\psi'\rangle &= \cos(\pi - \theta)|0\rangle + e^{i(\varphi + \pi)} \sin(\pi - \theta)|1\rangle \\
 &= -\cos\theta|0\rangle + e^{i\varphi} e^{i\pi} \sin\theta|1\rangle \\
 &= -\cos\theta|0\rangle + e^{i\varphi} \sin\theta|1\rangle \\
 &= -|\psi\rangle
 \end{aligned}$$

The Bloch sphere

which leads to

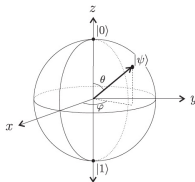
$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

where $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq 2\pi$



The map $\frac{\theta}{2} \mapsto \theta$ is **one-to-one** at any point but:
all points on the equator are mapped into a single point: the south pole.

The Bloch sphere



- The poles represent the classical bits. In general, **orthogonal states correspond to antipodal points** and every **diameter** to a **basis** for the single-qubit state space.
- Once measured a qubit collapses to one of the two poles. Which pole depends exactly on the arrow direction: The angle θ measures that **probability**: If the arrow points at the equator, there is 50-50 chance to collapse to any of the two poles.
- Rotating a vector wrt the z -axis results into a **phase change** (φ), and does not affect which state the arrow will collapse to, when measured.

End of parenthesis

...)